

ServiceNow-Based Employee Licensed Software Installation Application

Project Documentation

Platform: ServiceNow (Global Scope)
Module Type: Service Catalog / Catalog Item / Flow Designer
Team Members: [Add team member names and roles here]

1. Introduction

Project Title

ServiceNow-Based Employee Licensed Software Installation Application.

Team Members

[Add team member names and roles here]

Platform Overview

This project is built entirely on the ServiceNow platform using out-of-the-box (OOB) Service Catalog and Flow Designer capabilities in the Global application scope. Unlike a traditional MERN-stack build — and unlike a custom scoped application — the 'frontend', 'backend', and 'database' layers here are all native ServiceNow constructs: a Service Catalog Item for the UI, a Flow Designer flow for server-side approval/fulfillment logic, and standard ServiceNow tables (sc_req_item, sc_task, sysapproval_approver) for data storage — no custom table was required.

2. Project Overview

Purpose

The Licensed Software Installation Application allows employees to raise a request for licensed software (e.g. Microsoft Office, Adobe Acrobat, Visual Studio) directly from the Service Catalog. The request automatically routes to the Software Manager for approval, and — once approved — a fulfillment task is created for the IT Software Support Team. The goal is to digitize and standardize what would otherwise be an ad-hoc, email-based software request process, while keeping full visibility over license usage and installation status.

Key Features

- A Service Catalog Item ("Licensed Software Installation") that presents a structured request form to employees.
- Six request variables — Software Name, Software Version, Installation Device, License Type, Business Justification, and Required By Date — captured on the standard Requested Item (RITM) record.
- An approval workflow built in Flow Designer that asks the Software Manager to approve or reject each request.
- Conditional branching in the flow — fulfillment only proceeds when the approval state is Approved.
- Automatic Catalog Task creation ("Install Licensed Software") assigned to the Software Group / Software Manager on approval.
- Live status tracking on the Requested Item — Approval state, Stage, Quantity, and Delivery Plan.
- Centralized, auditable approval history on the Approvals list, filterable by approver.
- All configuration changes captured in a dedicated Update Set for safe promotion between instances.

3. Architecture

The application follows ServiceNow's standard three-layer model. The table below maps each MERN concept to its ServiceNow equivalent used in this project.

Layer	ServiceNow Equivalent
Frontend (React)	Service Catalog Item + native Order/Request form UI
Backend (Node.js / Express)	Flow Designer flow ("Licensed Software Installation - Approval and Fulfillment")
Database (MongoDB)	Standard tables: sc_req_item, sc_task, sysapproval_approver (no custom table)

3.1 Application Scope

All components (catalog item, variables, flow, and update sets) live in the Global application scope rather than a dedicated custom scoped application. This keeps the solution simple to build and deploy for a single, focused catalog process, though a custom scoped app (as used in larger implementations) would be a reasonable next step if the solution grows to include custom tables, roles, or business rules.

3.2 Data Model — Catalog Item Variables

Rather than a custom table, request data is captured through catalog item variables that populate the standard Requested Item (RITM) record. Each variable is defined with a Type and Order that controls how it renders on the request form.

Question (Variable)	Type	Order
Software Name	Select Box	100
Software Version	Single Line Text	200
Installation Device	Select Box	300
License Type	Select Box	400
Business Justification	Multi Line Text	500
Required By Date	Date	600

The screenshot displays the ServiceNow interface for the 'Catalog Item - Licensed Software Installation'. The top navigation bar includes links for All, Favorites, History, Workspaces, and Admin. Below the navigation bar, there are tabs for 'Variables (6)', 'Variable Sets', 'Catalog UI Policies', 'Catalog Client Scripts', 'Available For', 'Not Available For', 'Categories (1)', 'Catalogs (1)', 'Catalog Data Lookup Definitions', 'Related Articles', and 'Related Catalog Items'. The 'Variables (6)' tab is active, showing a table of variables. The table has columns for 'Type', 'Question', and 'Order'. The variables listed are: 'Software Name' (Select Box, Order 100), 'Software Version' (Single Line Text, Order 200), 'Installation Device' (Select Box, Order 300), 'License Type' (Select Box, Order 400), 'Business Justification' (Multi Line Text, Order 500), and 'Required By Date' (Date, Order 600). The bottom of the screen shows a message: 'No templates are available. Create A New One?'.

Type	Question	Order
Select Box	Software Name	100
Single Line Text	Software Version	200
Select Box	Installation Device	300
Select Box	License Type	400
Multi Line Text	Business Justification	500
Date	Required By Date	600

Figure 1: The 6 variables defined on the Licensed Software Installation catalog item.

3.3 Catalog Item — Employee Ordering Experience

The "Licensed Software Installation" Service Catalog Item is the ServiceNow equivalent of a public-facing intake form. It collects the software name, version, installation device, license type, business justification, and required-by date, and on submission creates a new Requested Item (RITM) record.

Employees can use this catalog item to request installation of licensed software on an approved company device. The request requires manager approval before the software installation task is assigned to the IT Software Support Team.

Exists in categories

Software Name
Microsoft Office

* Software Version
Microsoft Office 2024

* Installation Device
Laptop

* License Type
Per User

* Business Justification
this is for my project sake

* Required By Date
2026-07-29

Quantity: 1
Delivery time: 0 Days
Order Now
Add to Cart
Shopping Cart: Empty

Figure 2: Employee-facing catalog item request form.

3.4 Flow Designer — Approval and Fulfillment Workflow

The "Licensed Software Installation - Approval and Fulfillment" flow is triggered whenever a new item is ordered through the Service Catalog. It asks the Software Manager for approval, checks the approval outcome with an If condition, and — when approved — creates a Catalog Task and updates the Requested Item record.

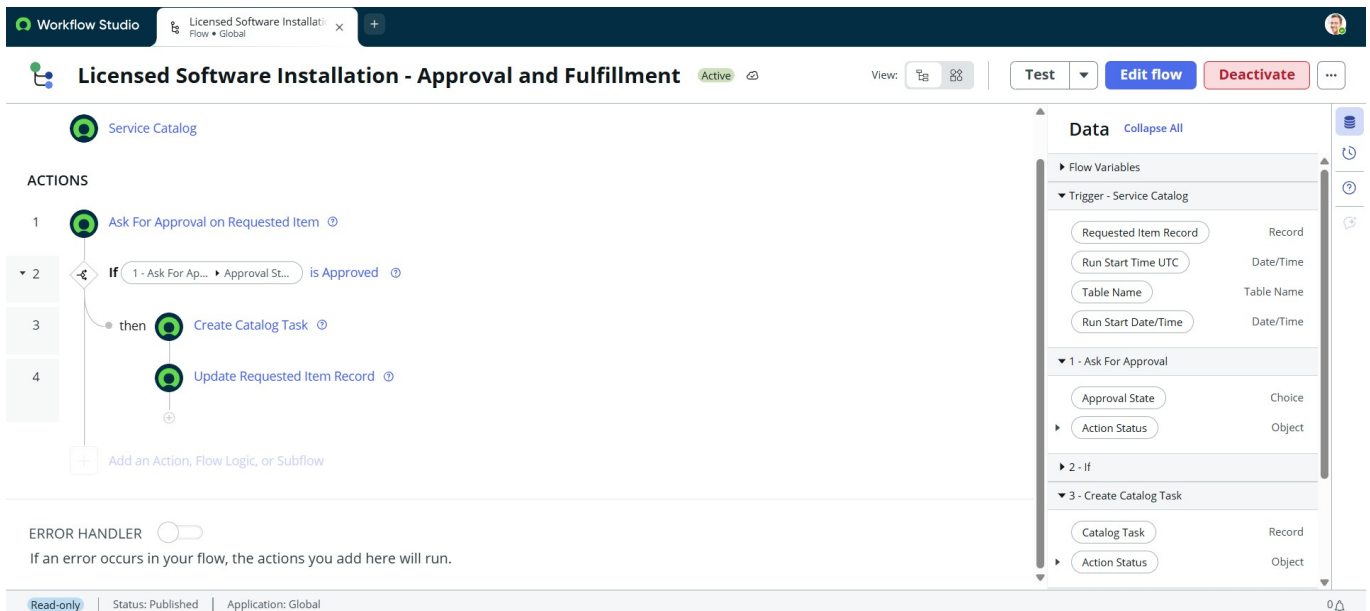


Figure 3: Flow overview — trigger (Service Catalog) and the four core actions.

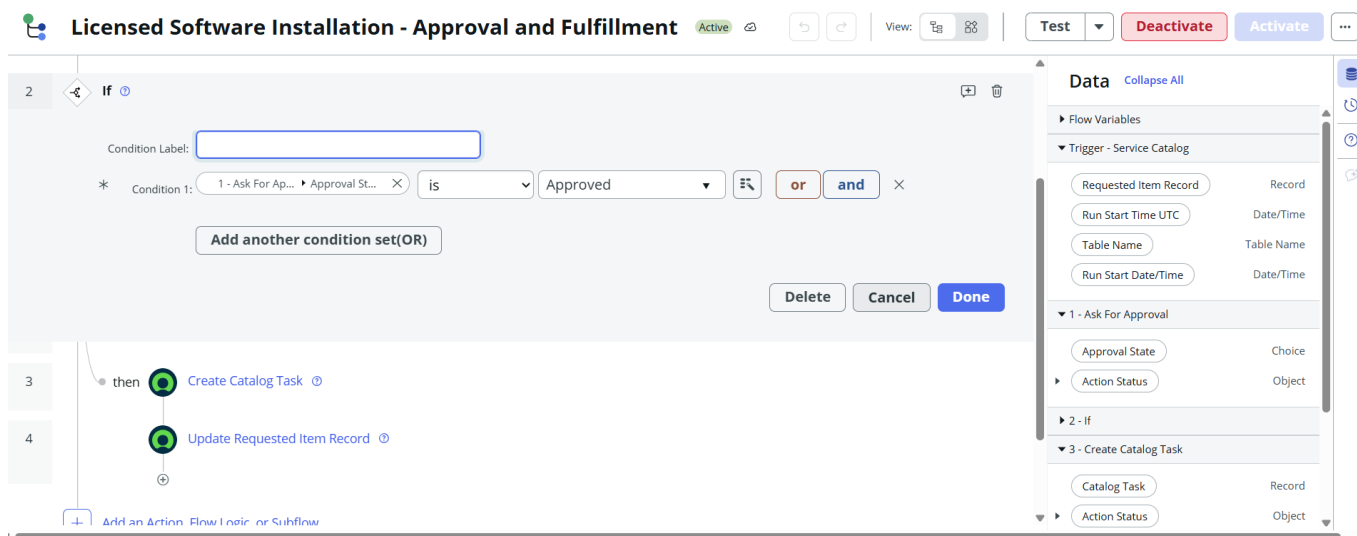


Figure 4: If-condition logic validating that the approval state is Approved.

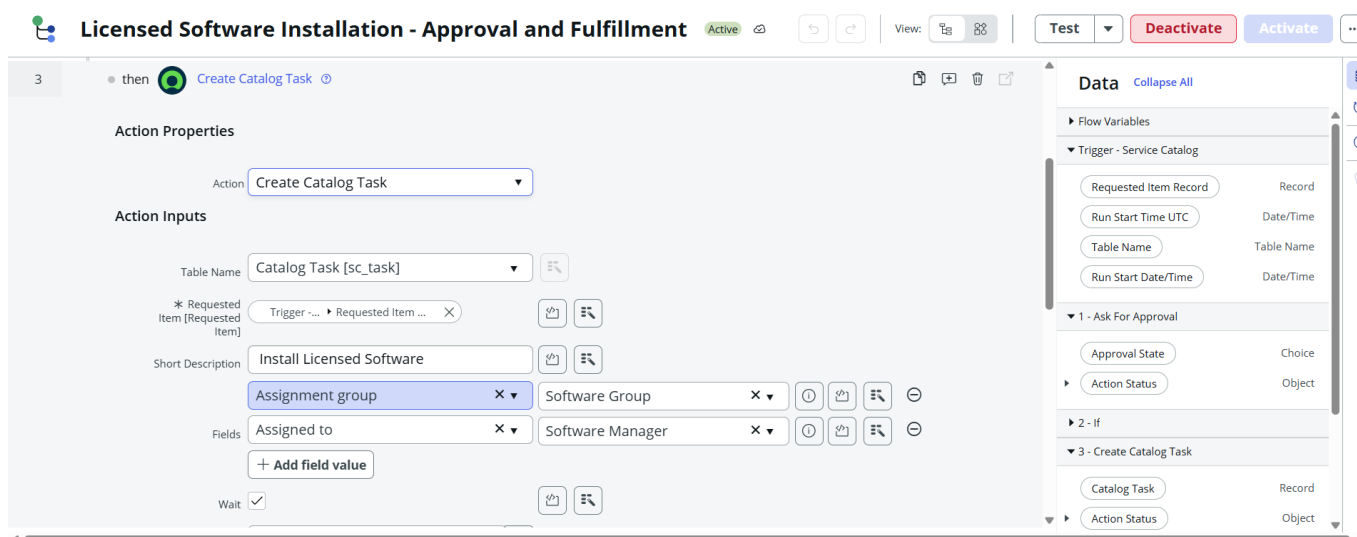


Figure 5: Create Catalog Task action — the fulfillment step, assigned to the Software Group / Software Manager.

4. Setup Instructions

Prerequisites

- A ServiceNow instance (Personal Developer Instance or sandbox) with admin access.
- The catalog_admin role, required to create or modify the catalog item and its variables.
- The flow_designer role, required to build and activate the approval/fulfillment flow.

Installation — Importing the Update Set

All customizations for this project (catalog item, variables, and flow) are captured across the "Licensed Software Installation" and "Licensed Software Installation - Development" update sets. To install the application on another instance:

- On the source instance, open the relevant Update Set and use Export to XML to download the remote update set XML file.
- On the target instance, navigate to Retrieved Update Sets and import the XML file.
- Open the imported Update Set and click Preview Update Set, resolve any conflicts, then click Commit Update Set.
- Verify that the Licensed Software Installation catalog item, its variables, and the approval/fulfillment flow have all been created and the flow shows as Active.

5. Application Structure

Instead of a client/server folder structure, this ServiceNow solution organizes its components by type. The main components created for this project are:

Component	Name / Table	Purpose
Catalog Item	Licensed Software Installation (sc_cat_item)	Employee-facing request form
Flow	Licensed Software Installation - Approval and Fulfillment	Approval routing and fulfillment logic
Requested Item	sc_req_item	Stores each employee's submitted request and its live status
Catalog Task	sc_task — "Install Licensed Software"	Fulfillment task for the IT Software Support Team
Approval Record	sysapproval_approver	Tracks each approval decision (Software Manager)
Update Sets	Licensed Software Installation / ...-Development	Configuration change tracking for deployment

Name	Application	State	Installed from	Created	Created by	Parent	Batch Base
Default	sravanth application	In progress		2026-06-22 10:33:15	admin	(empty)	(empty)
Default	(empty)	Ignore		2026-07-18 19:45:38	admin	(empty)	(empty)
Default	Benchmarks Spoke	In progress		2026-06-09 13:15:10	admin	(empty)	(empty)
Default	Global	In progress		2026-04-01 09:55:59	system	(empty)	(empty)
Default	Student Leave Management	In progress		2026-07-22 17:05:55	admin	(empty)	(empty)
Default	Software Installation Request	In progress		2026-07-24 19:13:02	admin	(empty)	(empty)
Default	Admin Center	In progress		2026-06-22 15:42:43	admin	(empty)	(empty)
ka	Global	Complete		2026-06-05 12:19:01	admin	(empty)	(empty)
Licensed Software Installation	Global	Complete		2026-07-24 10:17:25	admin	(empty)	(empty)
Licensed Software Installation - Develop...	Global	In progress		2026-07-27 16:01:50	admin	(empty)	(empty)
Network Request catalog item	Global	In progress		2026-07-25 19:08:19	admin	(empty)	(empty)
sravanth	Global	Complete		2026-06-05 12:20:51	admin	(empty)	(empty)

Figure 8: Update Sets list, including the Licensed Software Installation update sets.

Update Set
Licensed Software Installation - Development

* Name: Licensed Software Installation - Development

State: In progress

Parent:

Release date:

Install date:

Installed from:

Description: Update set for Licensed Software Installation Service Catalog project

Application: Global

Created: 2026-07-27 16:01:50

Created by: admin

Merged to:

Update

Related Links

[Merge With Another Update Set](#)

[Scan Update Set](#)

[\[SN Utils\] Versions \(0\)](#)

Customer Updates (22) | Update Set Logs | Child Update Sets | Install History

Created | Search

No templates are available. [Create A New One?](#)

Figure 9: Update Set record — scope, state, and description of the change set.

Update set = Licensed Software Installation - Development

Created	Type	View	Target name	Updated by	Remote update set	Action
2026-07-27 16:12:53	Variable		License Type	admin	(empty)	INSERT_OR_UPDATE
2026-07-27 16:10:08	Variable		Software Version	admin	(empty)	INSERT_OR_UPDATE
2026-07-27 16:06:49	Variable		Software Name	admin	(empty)	INSERT_OR_UPDATE
2026-07-27 16:14:47	Variable		Business Justification	admin	(empty)	INSERT_OR_UPDATE
2026-07-27 16:10:54	Variable		Installation Device	admin	(empty)	INSERT_OR_UPDATE
2026-07-27 16:16:30	Variable		Required By Date	admin	(empty)	INSERT_OR_UPDATE
2026-07-27 16:08:25	Question Choice		IntelliJ IDEA	admin	(empty)	INSERT_OR_UPDATE
2026-07-27 16:11:19	Question Choice		Laptop	admin	(empty)	INSERT_OR_UPDATE
2026-07-27 16:11:35	Question Choice		Desktop	admin	(empty)	INSERT_OR_UPDATE
2026-07-27 16:07:50	Question Choice		Visual Studio	admin	(empty)	INSERT_OR_UPDATE
2026-07-27 16:08:08	Question Choice		Adobe Acrobat	admin	(empty)	INSERT_OR_UPDATE
2026-07-27 16:07:18	Question Choice		Microsoft Office	admin	(empty)	INSERT_OR_UPDATE
2026-07-27 16:08:42	Question Choice		AutoCAD	admin	(empty)	INSERT_OR_UPDATE

No templates are available. [Create A New One?](#)

Figure 10: Customer Updates within the update set — every variable and choice value created for the catalog item.

6. Running the Application

There is no separate frontend/backend server to start — the application runs natively inside the ServiceNow instance.

- Employee: opens the "Licensed Software Installation" item from the Service Catalog, fills in the form, and clicks Order Now.
- Software Manager: approves or rejects the request from the Approvals list or the approval notification.
- IT Software Support Team: works the generated Catalog Task once the request is approved.

Once submitted, the Requested Item (RITM) is created and the flow automatically raises an approval. After approval, the RITM shows the live Delivery Plan status for the "Install Licensed Software" task.

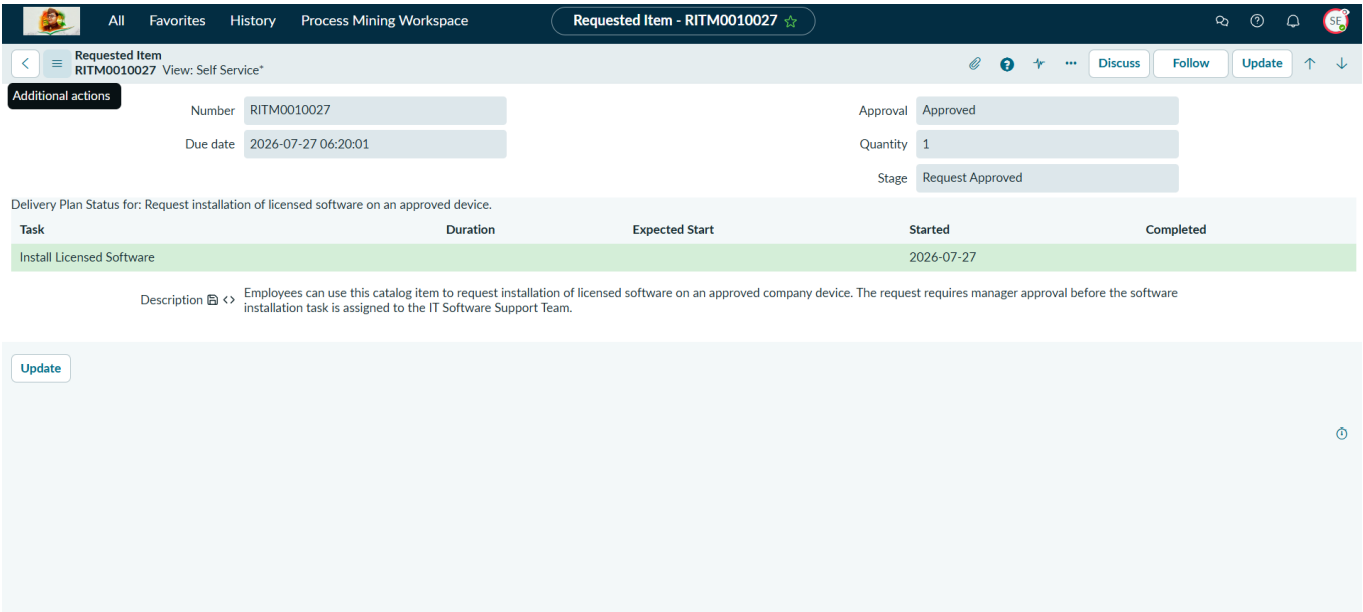


Figure 7: RITM0010027 after manager approval, showing the Install Licensed Software delivery task in progress.

7. Data / API Documentation

ServiceNow does not require hand-written REST endpoints for basic CRUD — every table automatically exposes a REST API (the Table API). The table below documents the operations available and the equivalent UI action that triggers them in this application.

Operation	Table API Method	Triggered From
Create a request	POST /api/now/table/sc_req_item (via Catalog API)	Catalog item order submission
Read a request	GET /api/now/table/sc_req_item/{sys_id}	Opening the Requested Item (RITM) record
Update a request	PUT /api/now/table/sc_req_item/{sys_id}	Flow's "Update Requested Item Record" action
Create fulfillment task	POST /api/now/table/sc_task	Flow's "Create Catalog Task" action
List approvals	GET /api/now/table/sysapproval_approver	Approvals list, filtered by approver

Access to every one of these operations is still governed by ServiceNow's standard Access Control Rules (ACLs) — the Table API enforces the same role checks as the UI.

8. Authentication & Authorization

Instead of JWTs or sessions managed in application code, this project relies entirely on ServiceNow's built-in platform security: native user sessions and standard, out-of-the-box roles rather than custom roles, groups, or ACLs.

8.1 Roles

- `catalog_admin` — required to create or modify the catalog item and its variables.
- `flow_designer` — required to build, edit, and activate the approval/fulfillment flow.
- `approver_user` — allows the Software Manager to act on approval requests routed to them.
- `itil` — used by the IT Software Support Team to work the generated Catalog Task.

8.2 Access Control

Read/write access to `sc_req_item`, `sc_task`, and `sysapproval_approver` is governed by ServiceNow's standard, out-of-the-box Access Control Rules for those tables. No custom roles, groups, or ACLs were created for this iteration of the project — this is called out as a Future Enhancement below for tighter, request-specific security.

9. User Interface

The end-user interface is entirely native ServiceNow UI — no custom frontend code was written. Two surfaces make up the user experience:

- Employee-facing: the "Licensed Software Installation" Service Catalog Item form, used to submit a new request.
- Manager/admin-facing: the standard ServiceNow Approvals list and the Requested Item (RITM) record, which shows the request details together with its live Delivery Plan status.

Because the form is built with out-of-the-box ServiceNow UI elements (select boxes, single/multi-line text, and a date picker), it automatically inherits the platform's responsive layout, accessibility, and theming.

10. Testing

Testing for this application was carried out manually within the ServiceNow instance, using the following approach:

- Form submission testing: submitting the Licensed Software Installation form with valid and missing field values to confirm mandatory fields (marked with *) are enforced.
- Approval routing testing: confirming the flow correctly routes the request to the Software Manager and that Approved vs Rejected outcomes branch the flow as expected.
- Fulfillment testing: confirming the Create Catalog Task action only fires on approval, with the correct Assignment Group (Software Group) and Assigned To (Software Manager) values.
- Status tracking testing: verifying the RITM's Stage and Delivery Plan status update correctly as the Catalog Task progresses.
- Update Set validation: exporting the update set to XML and re-importing it into a clean instance to confirm the catalog item, variables, and flow all migrate correctly.

11. Screenshots / Demo Summary

Screenshots for every major component are embedded throughout this document, in the section that describes the corresponding feature (Architecture, Setup, Running the Application, and Data/API Documentation). The quick index below lists every figure for reference.

Fig.	Shows
1	Catalog item variables (Software Name, Version, Device, License Type, Justification, Date)
2	Employee-facing catalog item request form
3	Flow trigger and overview of the four core actions
4	If-condition validating the approval state is Approved
5	Create Catalog Task action — the fulfillment step
6	Approvals list showing decisions across multiple software requests
7	RITM after manager approval, showing the delivery task in progress
8	Update Sets list, including the Licensed Software Installation update sets
9	Update Set record — scope, state, and description
10	Customer Updates within the update set

State	Approver	Comments	Approval for	Created
No Longer Required	Software Manager		RITM0010024	2026-07-27 05:42:54
Approved	Software Manager		RITM0010024	2026-07-27 05:43:12
Rejected	Software Manager	2026-07-27 06:00:39 - Software Manager (	RITM0010025	2026-07-27 05:57:48
Approved	Software Manager		RITM0010027	2026-07-27 06:20:04
Rejected	Software Manager	2026-07-27 06:30:05 - Software Manager (	RITM0010028	2026-07-27 06:29:22

Figure 6: Approvals list showing decisions across multiple software requests.

12. Known Issues

- Software Name and Software Version are free-text/select values without a backing CMDB or Software Model lookup, which can lead to inconsistent naming across requests.
- There is no automated license/seat-availability check before approval — Software Asset Management (SAM) is not yet integrated into the flow.
- All components currently live in the Global scope with standard OOB roles; there are no request-specific custom roles, groups, or ACLs restricting access the way a dedicated scoped app would.
- Rejection reason is captured as free-text comments on the approval record rather than a structured reason code.

13. Future Enhancements

- Integrate Software Asset Management (SAM) to validate license seat availability automatically before approval.
- Add automated fulfillment via SCCM/Intune/JAMF integration so the Catalog Task triggers a real software push instead of a manual IT task.
- Introduce custom roles/groups and request-specific ACLs (similar to an "it employees"/"it manager" model) for tighter access control.
- Add a structured rejection-reason field and an automatic re-submission workflow.
- Build a Performance Analytics / reporting dashboard summarizing requests by software title, status, and turnaround time.
- Add mobile-friendly UI Actions so managers can approve or reject requests directly from the ServiceNow mobile app.